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Chromatography of Poliovirus on Calcium Phosphate and Its Application to the Identification of Vaccine Progeny Strains

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ABSTRACT

OZAKI, YOSHIKATSU (Baylor University College of Medicine, Houston, Tex.), ARWIN R. DIWAN, MIZUHO TAKIZAWA, AND JOSEPH L. MELNICK. Chromatography of poliovirus on calcium phosphate and its application to the identification of vaccine progeny strains. J. Bacteriol. **89**:603–610. 1965.—Virulent and attenuated polioviruses were studied chromatographically with calcium phosphate as adsorbent and neutral phosphate buffer as eluent. For type 1, the attenuated LSc2ab strain and the virulent Mahoney strain were eluted at different salt concentrations. The peak of elution for the LSc2ab strain was at 0.3 M, whereas for the Mahoney strain it was at 0.05 M. No differences in elution pattern were noted with virulent and attenuated strains of poliovirus types 2 and 3. Except for a single instance, progeny strains isolated from vaccines or contacts 1 to 8 weeks after vaccination showed an elution pattern similar to the parent vaccine strain, or intermediate between the virulent and attenuated strains but still clearly distinguishable from wild-type virus. A virus population that possessed the same elution property as the virulent Mahoney strain could be separated from a number of progeny strains by chromatography. The ratio of this easily eluted population appeared to increase during the course of multiplication of vaccine virus in the human intestine. A virus population with the same elution profile as Mahoney seems to be present in the current LSc2ab oral polio vaccine, but as a very minor component. However, this component could readily be enriched by selecting for passage only those virus particles eluted at low salt concentrations. Heat-resistant, attenuated virus strains were selected from virulent type 1 strains by the Wallis-Melnick procedure of repetitive heating of progeny in $AlCl_3$. However, the elution pattern remained the same as that of the parent virulent virus.

The adsorptive capacity of various adsorbents, such as aluminum phosphate, calcium phosphate, and modified cellulose, has been used widely for the purification and concentration of viruses. More recently, the elution pattern of virulent strains of poliovirus type 1 on diethylaminoethyl-(DEAE)-cellulose columns (Hodes, Zepp, and Ainbender, 1960) and on $Al(OH)_3$ gels (Woods and Robbins, 1961) was shown to be different from that of type 1 attenuated strains.

Calcium phosphate chromatography has been used by Taverne, Marshall, and Fulton (1958) as a simple and useful method for the purification and concentration of influenza virus particles and soluble antigens. The method has since been applied to the arboviruses (Smith and Holt, 1961), adenoviruses (Simon, 1962), and, most recently, polioviruses (Koza, 1963). The present paper describes the results obtained by chromatography of known virulent and attenuated strains of poliovirus on calcium phosphate columns, and the use of such columns for selecting more uniform virus populations. In addition, vaccine progeny in the form of virus recovered from persons receiving type 1 vaccine (LSc2ab) and from their contacts has been studied to evaluate the method for monitoring oral polio vaccine. Comparative results with other in vitro markers, particularly the antigenic marker, are also presented.

MATERIALS AND METHODS

Tissue culture. Primary rhesus monkey kidney (MK) cells and the FL continuous cell line were used for virus growth and assay. Melnick's medium, containing 0.2 mM $AlCl_3$ to suppress ad-

ventitious agents (Wallis and Melnick, 1962), was used as MK cell nutrient, and the same medium, containing 0.1% yeast extract and 8% calf serum, was used for the FL cells.

Viruses. Virulent strains were plaque-purified lines of poliovirus type 1, Mahoney; type 2, MEF1; type 3, P24; and the unpurified Brunhilde type 1 strain. The attenuated strains were those used in the oral polio vaccine (type 1, LSc2ab; type 2, P712; and type 3, Leon). Also used in the study were strains recovered at weekly intervals from vaccinees. (The Japanese strains were kindly supplied by I. Tagaya and A. Kono, National Institute of Health, Tokyo, Japan. The Houston strains were from vaccinees I-33, I-12, and I-37, kindly supplied by Matilda Benyesh-Melnick.) They were used in their first passage in tissue culture. Other attenuated strains, selected by the Wallis-Melnick (1963) method of heating virulent virus in the presence of $AlCl_3$, were also studied. All virus assays were carried out by determination of plaque-forming units (PFU) in 1-oz bottles by methods described (Wallis, Melnick, and Bianchi, 1962).

Preparation of calcium phosphate. Analytical grade reagents were used. The calcium phosphate was prepared by slowly adding an equal volume of 0.5 M $CaCl_2$ to 0.5 M Na_2HPO_4, which was constantly stirred by means of a magnetic mixer. The resultant precipitate was washed three or four times and stored at 4 C in 0.001 M phosphate buffer (*p*H 7.0).

TABLE 1. *Recovery of attenuated and virulent type 1 poliovirus from columns*

Fraction	LSc, 2ab (attenuated; d− T− A−)*		Mahoney (virulent; d+ T+ A+)	
	PFU/ml	Per cent recovery	PFU/ml	Per cent recovery
Original virus fluid	4.5×10^7	—	1.8×10^8	—
Virus fluid after passage through column	1.0×10^2	—	1.0×10^6	—
F1 (0.05 M)†	1.6×10^3	0.003	1.4×10^8	*78.0*
F2 (0.1 M)	1.4×10^4	0.03	6.5×10^6	3.6
F3 (0.2 M)	6.5×10^4	0.1	2.5×10^5	0.14
F4 (0.3 M)	2.0×10^7	*45.0*	8.0×10^4	0.05
F5 (0.4 M)	7.5×10^5	1.7	4.0×10^4	0.02
F6 (0.5 M)	1.3×10^5	0.2	2.2×10^4	0.01

* Symbols: d+ = high efficiency of plating under low bicarbonate agar, d− = low efficiency of plating under low bicarbonate agar; T+ = efficient growth of virus at 40 C, T− = almost no growth at 40 C; A+ = inactivation of virus at 50 C in 10 mM $AlCl_3$, A− = stabilization of virus at 50 C in 10 mM $AlCl_3$.

† F1 = fraction 1, F2 = fraction 2, etc.

Phosphate buffer. Stock 1 M neutral phosphate buffer was prepared with 0.66 M Na_2HPO_4 and 0.33 M KH_2PO_4. It was diluted as needed.

Preparation of the column. Calcium phosphate suspension (10 ml) was added to a glass column (10 by 500 mm) and packed under about 2 psi of nitrogen pressure. The columnar volume was measured by the following method. A drop of bromocresol purple solution was placed on the surface of the calcium phosphate, distilled water was carefully run through the column under nitrogen pressure, and the amount of eluate collected before the first drop of dye appeared was regarded as the volume of the column.

Operation of the column. When the column was ready for use, one columnar volume of undiluted virus suspension was added. Slight positive pressure was applied, so that the virus suspension replaced the water which was in the column, and the virus adsorbed to the column. Subsequently, elution at *p*H 7.0 was carried out with a continuous gradient of 0.05 to 0.5 M phosphate buffer. One columnar volume of each buffer was passed in series through the column; the effluents were labeled as M (virus remaining in medium after adsorption) and fractions 1 to 6, ranging from low to high salt concentrations.

To test the effect of volume of eluate, after virus adsorption the column was washed with double volumes of the buffer at each molarity. The first fractions and the washings were collected and titrated separately. The results obtained showed that the virus populations were eluted at exactly the same molarities as the fractions obtained without additional washings. Furthermore, a population eluted at a given molarity, when refractionated on another column, could be eluted at precisely the same molarity. The column was cleaned by using concentrated HCl, which dissolves the calcium phosphate. The tube was then rinsed with water, washed, cleaned, and autoclaved.

Antigenic analysis. The antigenic character analysis was carried out by studying serum neutralization kinetics by use of methods previously described (Ozaki et al., 1963). Rabbit antiserum to the LSc strain was used.

Other in vitro markers. The d, T, and A markers (see Table 1) were studied with the methods previously described in reports from the Baylor laboratory (Benyesh-Melnick and Melnick, 1959; Wallis et al., 1962).

RESULTS

Elution of virulent and attenuated type 1 polioviruses. Table 1 shows the results of a typical experiment carried out in Houston with the virulent Mahoney strain and the attenuated LSc strain. There was no significant difference in adsorption of these viruses: 99% of Mahoney virus, and nearly 100% of LSc virus, were adsorbed. A large amount (78%) of Mahoney virus

was eluted in fraction 1 (0.05 M), whereas almost no LSc virus was eluted in the first three fractions; 45% of LSc virus was obtained in fraction 4 (0.3 M), but little Mahoney virus was recovered in the later fractions, 2 through 6.

Figure 1 illustrates a typical experiment carried out in Kyoto with the virulent Mahoney and Brunhilde strains and the attenuated LSc strain. There was no significant difference in the behavior of the virulent strains. Both had peaks in fraction 1, which was eluted from the column with 0.05 M phosphate buffer solution; more than 90% of input virus was recovered in this fraction. Relatively small amounts of the virulent strains were found in subsequent fractions. In contrast, the LSc peak (over 90% recovery) was found in fraction 4 (0.3 M), and relatively little was eluted in fractions 1 to 3. The results of a number of experiments conducted in both laboratories repeatedly showed that the virulent strains were eluted at low ionic strength (0.05 M), whereas the attenuated strain was not eluted until the salt concentration reached 0.3 M. This finding indicated that this chromatographic procedure could easily distinguish virulent polioviruses of the Mahoney and Brunhilde types from attenuated virus of the LSc type. Therefore, the method was applied to LSc progeny strains to determine the constancy of the marker.

Progeny strains recovered from vaccinees and their contacts. The elution pattern of three strains

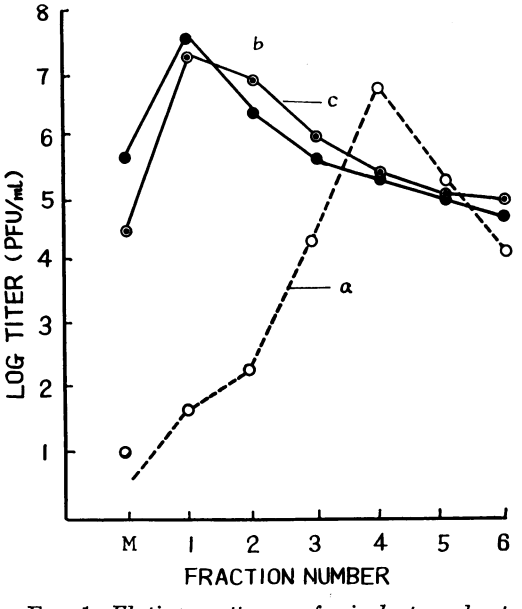


FIG. 1. *Elution patterns of virulent and attenuated type 1 poliovirus strains. (a) LSc strain, (b) Mahoney strain, (c) Brunhilde strain.*

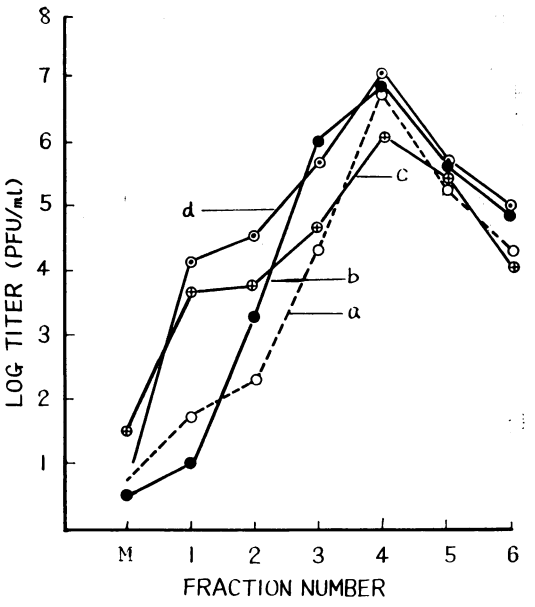


FIG. 2. *Elution patterns of progeny strains isolated from a contact (C-1) of a vaccinated child. (a) LSc vaccine strain, (b) C-1-1 strain isolated 1 week after vaccination, (c) C-1-7 strain isolated 7 weeks after vaccination, (d) C-1-9 strain isolated 9 weeks after vaccination.*

isolated from a Tokyo vaccinee, V-1, showed that the sample obtained 1 week after oral vaccination had a pattern similar to the vaccine strain shown in Fig. 1. The 5th- and 8th-week strains, however, had 100 times more virus eluted in fraction 2 than did the parent strain.

Figure 2 shows the results obtained with isolates from a vaccine contact, C-1. Although the elution pattern of the strain obtained 1 week after infection was very much like that of LSc, the 7th- and 9th-week strains again showed a shift to the left. More than 1,000 times as much virus was found in fraction 1 of the 9th-week strain as was found in fraction 1 of the vaccine strain. However, it was found that isolates from another contact, C-2, showed the same elution peak (fraction 4) as did LSc—regardless of the time of isolation—and precisely the same outline at all salt concentrations. Some progeny strains demonstrated even greater differences from the vaccine strain than did those shown in Fig. 2. Thus, Fig. 3 illustrates the elution patterns of two strains isolated from vaccinee V-2 and contact C-3, 4 and 7 weeks, respectively, after the vaccine feeding. These two strains show a pattern intermediate between the virulent and attenuated strains.

Table 2 shows the results of tests with three

type 1 strains recovered from Houston vaccinee I-33 at biweekly intervals. The 1st and 3rd week postvaccination specimens showed a pattern of elution similar to that of LSc, the vaccine strain. By the 5th week, however, the virus had changed, and it now was eluted at low salt concentration, as was Mahoney virus. Other Houston type 1 vaccinees did not show as marked shifts. Vaccinee I-12 yielded a first week postvaccination specimen that behaved like vaccine virus, but, by the 4th week, virus was appearing in fraction 1—the fraction in which Mahoney virus appears—as well as in fractions 2 and 3. Strains recovered through the 5th postvaccination week from another Houston type 1 vaccinee, I-37, behaved like vaccine virus.

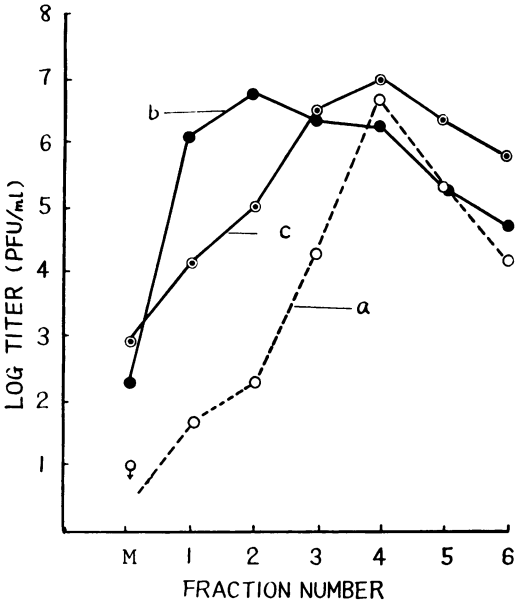


FIG. 3. *Elution patterns of progeny strains isolated from a vaccinated person, V-2, and a contact, C-3. (a) LSc vaccine strain, (b) V-2-4 strain isolated 4 weeks after vaccination, (c) C-3-7 strain isolated 7 weeks after vaccination.*

Separation of virus populations. The above experiments show that the elution property of some progeny viruses tends to deviate from the pattern of the vaccine virus toward that of virulent viruses. If a virus strain is not homogeneous, variation in the elution patterns of its progeny might be due to differences in ratio of the several kinds of particles. If this were true, a homogeneous population should be separable from the mixed population by collecting and testing various fractions obtained by chromatography. To test this hypothesis, the following experiments were performed with progeny of strains isolated from a Tokyo vaccinee, V-1, and from two contacts, C-1 and C-3. The recovered virus in fraction 1 was cycled once more through a column, and the virus in the new fraction 1 was passed in tissue culture. The harvested virus was then passed through a column to determine its elution pattern.

As seen in Fig. 4, the elution property of the F1 subpopulation after a single passage of strains V-1-5 and V-1-8 was almost identical to that of the virulent Mahoney virus. With the other progeny strains isolated at 7 weeks from contacts C-1 and C-3, the results were similar. These findings suggest that the original LSc2ab vaccine stock, even though derived by Sabin (1957) from a single plaque, is still heterogeneous. The deviation in the elution pattern from that of the original LSc vaccine virus might be due to a change in the ratio of two kinds of virus particles, which probably had occurred by selection during

TABLE 2. *Elution patterns of type 1 poliovirus strains recovered from Houston vaccinee I-33**

Fraction	Strains recovered after vaccination					
	1st week (d ± T − A−)†		3rd week (d − T − A−)		5th week (d + T − A+)	
	PFU/ml	Per cent recovery	PFU/ml	Per cent recovery	PFU/ml	Per cent recovery
Original	2.0×10^7	—	2.5×10^7	—	2.4×10^7	—
F1 (0.5 M)	1.0×10^3	0.005	$<1.0 \times 10^2$	—	1.5×10^7	*62.5*
F2 (0.1 M)	1.8×10^3	0.009	5.0×10^3	0.02	1.0×10^6	4.0
F3 (0.2 M)	8.0×10^4	0.4	1.0×10^6	4.0	6.5×10^5	2.0
F4 (0.3 M)	1.0×10^7	*50.0*	1.0×10^7	*40.0*	2.3×10^4	0.09
F5 (0.4 M)	1.7×10^5	0.9	2.5×10^6	10.0	7.0×10^3	0.03
F6 (0.5 M)	1.0×10^4	0.05	2.5×10^5	1.0	3.5×10^3	0.01

* Results for LSc vaccine and Mahoney strains are presented in Table 1.
† See footnote to Table 1 for definition of d, T, and A markers.

the course of virus multiplication in the human intestine. If the change were due to selection, virus possessing the elution property of virulent virus should be demonstrable as a minor population mixed with the major population of the vaccine virus itself.

To test this hypothesis, a separation of such a minor fraction was attempted from the LSc2ab vaccine virus. The vaccine was fractionated on the column, the virus recovered in fraction 1 was passed six times through new columns, and a tissue culture passage was made. With serial column passage, a shift of the elution pattern to the left became marked. After six passages, the virus had an elution peak in fractions 1 to 2 instead of in fraction 4, as in the original virus. This new elution property was stable during the repeated rapid passage of the substrain in tissue culture.

Comparison of elution property with antigenicity. Antigenicity of 19 progeny strains was investigated by the method of neutralization kinetics (Ozaki et al., 1963) to determine whether alteration of the elution property in the vaccine progeny during the course of multiplication in the human intestine was accompanied by a change in antigenic character. The experiments were set up with use of antiserum against the LSc2ab virus. At 1 week after vaccination, a number of isolates yielded normalized K values of 90 to 100, indicating antigenic identity with the vaccine virus, but the strains isolated at 4 to 9 weeks after feeding had shifted antigenically. The observation of this change in the virus during its multiplication in the human alimentary tract confirms earlier studies (Woods, Weiss, and Robbins, 1962; Wasserman and Fox, 1962; Gelfand, Nakano, and Cole, 1962; Ozaki et al., 1963). Table 3 shows that a simultaneous shift of the antigenic character and the elution marker does not occur. In these studies, 12 of 19 strains demonstrated an antigenic shift away from the vaccine strain, yet 8 of these still possessed the elution pattern of the vaccine virus and only 1 had a truly Mahoney-like pattern. Thus, the elution marker seems more stable than the antigenic marker.

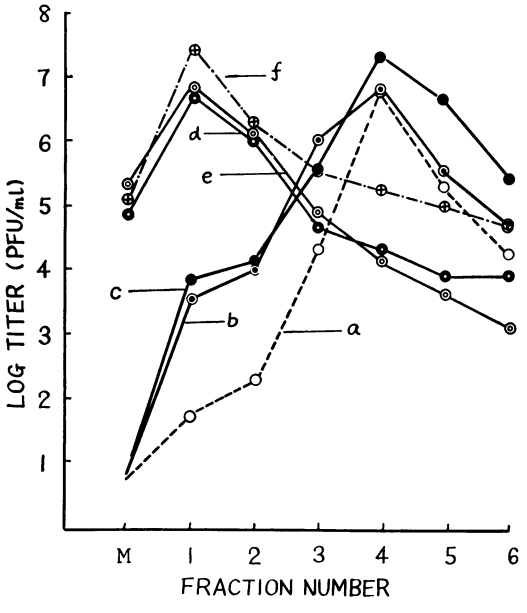


FIG. 4. *Elution patterns of substrains obtained from fraction 1 of progeny strains, V-1-5 and V-1-8. (a) LSc vaccine strain, (b) V-1-5 strain, (c) V-1-8 strain, (d) substrain obtained by passage of fraction 1, V-1-5 strain, (e) substrain obtained by passage of fraction 1, V-1-8 strain, (f) Mahoney strain.*

TABLE 3. *Summary of 19 progeny strains tested for antigenicity and chromatography markers*

Antigenicity	No. of strains	$Ca_3(PO_4)_2$ elution pattern		
		Vaccine-like	Intermediate	Mahoney-like
Vaccine-like*	7	7	0	0
Shift from vaccine†	12	8	3	1

* Normalized K values over 80 (*see* Ozaki et al., 1963).

† Normalized K values under 80.

Chromatography of $AlCl_3$-attenuated strains. Wallis and Melnick (1963) demonstrated that attenuated poliovirus strains can be selected by heating virulent Mahoney virus (d+, T+, A+) in the presence of 10 to 100 mM $AlCl_3$ at 50 C. Survivors of each treatment were passed in cell cultures, and the resultant harvests were likewise processed. The 11th passage obtained in the original Wallis-Melnick experiments was used. The virus had the following properties: d+, T−, A−, heat-resistant, and attenuated for monkeys. Figure 5 illustrates the elution pattern obtained for this strain (11y) compared with those of Mahoney and LSc. The pattern was similar to the original Mahoney, even though its neurovirulence activity was of the same order as that of the LSc strain (Wallis and Melnick, 1963).

Further experiments were carried out with variants derived in Kyoto from Mahoney and Brunhilde strains by the Wallis-Melnick procedure. Table 4 shows the difference in heat sensi-

tivity between the original viruses and the selected ones. The original Mahoney and Brunhilde strains were very sensitive to heating in $AlCl_3$ at 50 C for 15 min, but this sensitivity

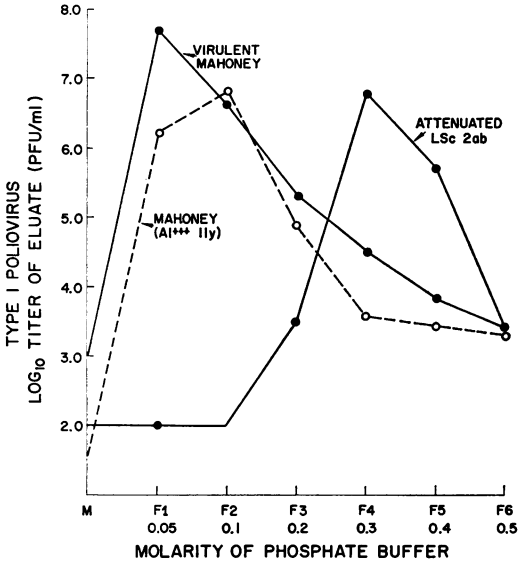


FIG. 5. *Elution pattern of type 1 poliovirus, strain 11y, derived from Mahoney strain by serial passage in culture, heating at each passage in the presence of 10 to 100 mM $AlCl_3$. Patterns of virulent Mahoney and attenuated LSc strains are shown for comparison.*

TABLE 4. *Selection by the Wallis-Melnick method of thermoresistant virus from the thermosensitive virulent strains*

Virus	Passage after serial $AlCl_3$ treatment	Virus titer (log PFU/ml)		Log difference
		Unheated sample	Heated* sample	
Mahoney	Original	7.5	2.5	5.0
	1	7.5	3.6	3.9
	2	7.5	4.0	3.5
	3	7.6	4.7	2.9
	4	7.6	6.4	1.2
	5	7.7	7.4	0.3
Brunhilde	Original	7.2	3.5	3.7
	1	8.0	5.2	2.8
	2	8.0	6.0	2.0
	3	7.6	7.1	0.5
	4	7.9	7.5	0.4
	5	7.6	7.3	0.3

* At 50 C for 15 min in presence of 12.5 mM $AlCl_3$.

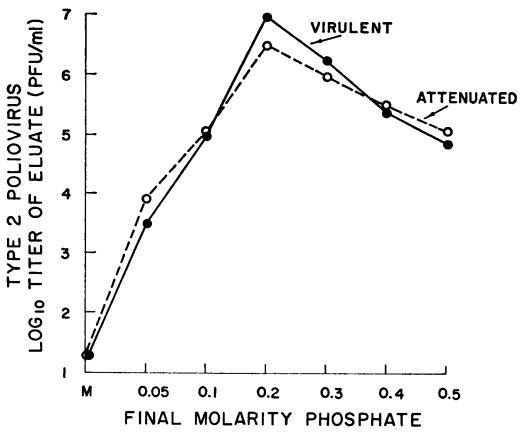


FIG. 6. *Elution patterns of virulent and attenuated strains of type 2 poliovirus.*

gradually decreased on repetition of the treatment at each passage. After five treatments, they were as resistant as the attenuated LSc virus. In spite of such a marked alteration of the Al marker, these modified strains did not differ in the elution property from the original virulent viruses. The experiments in both laboratories indicate that there is no correlation between the elution property and the Al marker.

Elution patterns of types 2 and 3. No significant difference was noted in the elution pattern of virulent and attenuated type 2 poliovirus (Fig. 6). The peak of elution was located in fraction 3 (0.2 M), different from that for type 1 strains. Virulent and attenuated type 3 strains yielded patterns like those of type 2. Four strains derived from the type 3 vaccine strain were tested. All showed the same elution pattern as the vaccine strain, even though two had the d− marker of the parent strain, whereas two had reverted to the wild d+ marker.

DISCUSSION

The elution patterns of several laboratory strains and a few progeny strains recovered from vaccinated persons have recently been reported (Hodes et al., 1960; Hollinshead, 1960; Delsal, Lepine, and Sautter, 1960; Woods and Robbins, 1961; Agol et al., 1962; Koza, 1963; Haas, 1963; Boeye, 1963). The results obtained with the calcium phosphate chromatography of type 1 strains are in agreement with the adsorption and elution behavior originally obtained with DEAE cellulose by Hodes et al. (1960) and with $Al(OH)_3$ by Woods and Robbins (1961); i.e., at low ionic strength and neutral *p*H, virulent strains are eluted more readily than attenuated variants.

After the LSc strain multiplies in vaccinated persons, population changes occur, as reflected in the elution pattern.

Virus recovered immediately after oral vaccination with the type 1 LSc2ab strain possesses an elution pattern similar to the vaccine strain. As multiplication continues in the body, the excreted virus often undergoes a pronounced change in its elution pattern, with peak virus appearing in the fraction of lower ionic strength than that required for LSc. In only one instance, however, did the progeny virus change so drastically as to be eluted as readily as was virulent Mahoney virus.

The antigenic character of poliovirus had been considered to be the most stable genetic marker, especially in comparison with the d, T, MS, and A markers, and it was hoped that it might be applied for the positive identification of progeny viruses isolated after the administration of vaccine. Thus, it has proved useful in the monitoring of certain stable type 1 and type 3 strains (Plotkin, Cohen, and Koprowski, 1961; Diwan et al., 1963; Ozaki et al., 1963). However, as recently shown (Woods et al., 1962; Wasserman and Fox, 1962; Gelfand et al., 1962), and as emphasized again in the present paper, the antigenic marker is not stable for type 1 LSc vaccine virus. Thus, progeny viruses may be difficult to differentiate from wild viruses by the use of any of the above markers. Although a change was also found to occur in the elution property of some LSc type 1 progeny isolates, it was not as marked as that seen in the antigenic character and other markers. Even when the progeny strains changed, they were almost always distinguishable from virulent type 1 viruses by their chromatographic behavior on calcium phosphate columns. This elution property may become the best available method for differentiation of LSc type 1 progeny from wild virus.

The shift in the elution pattern of the vaccine progeny strain toward that of virulent virus suggested that a mixed population was present in the progeny. This was proven by the separation of a fraction that had the same elution property as the virulent virus. It was further found that the LSc2ab vaccine stock, as currently used, is heterogeneous, for it still contains a minor component of virus particles with the elution pattern of the Mahoney strain. By chromatographic selection pressures, this fraction could be increased so that it became a major component of the population.

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